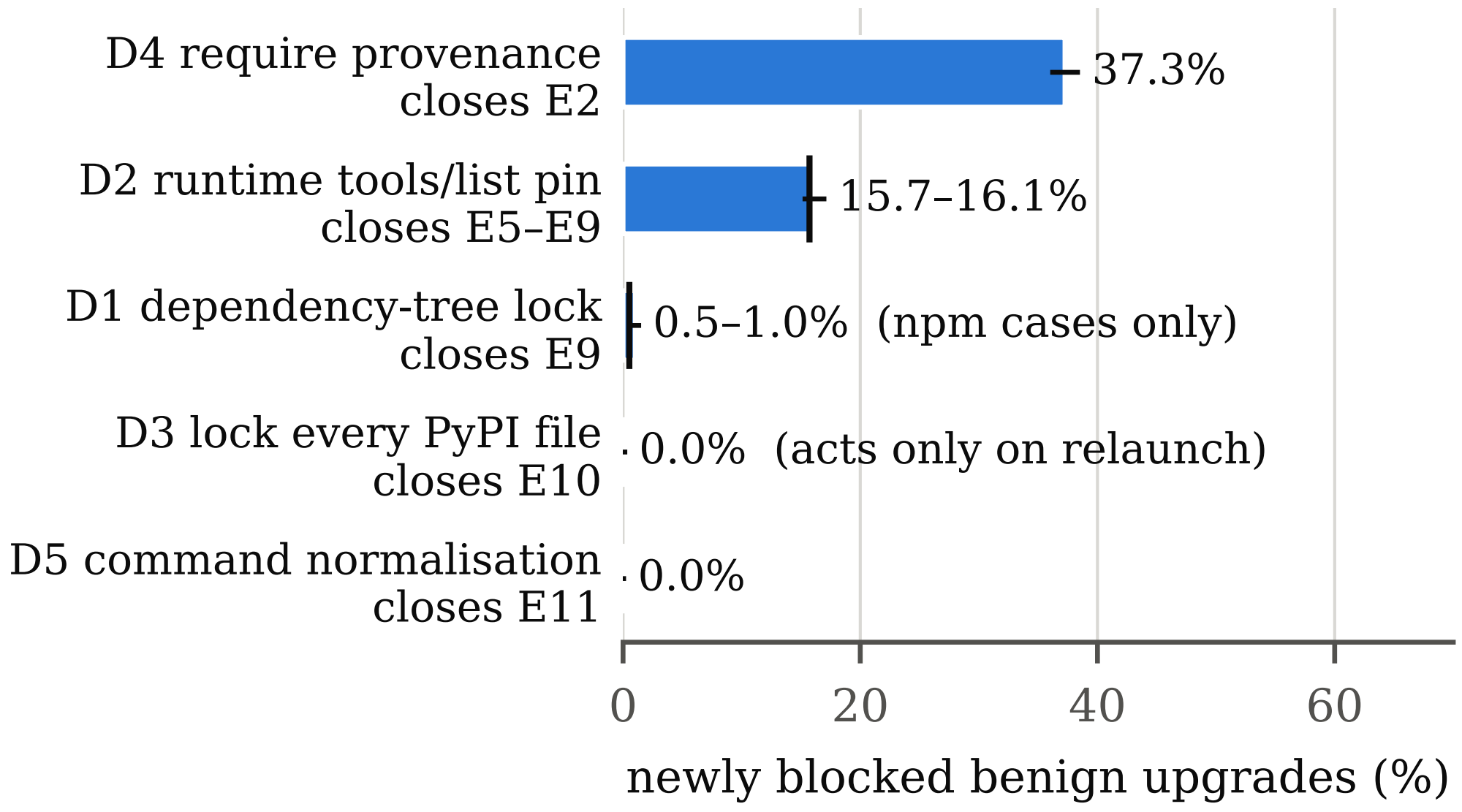
D4 require provenance closes E2
37.3%
D2 runtime tools/list pin closes E5–E9
15.7–16.1%
D1 dependency-tree lock closes E9
0.5–1.0% (npm cases only)
D3 lock every PyPI file closes E10
0.0% (acts only on relaunch)
D5 command normalisation closes E11
0.0%
0
20
40
60
newly blocked benign upgrades (%)